

Domain-Level Evolutionary Constraint Governs TP53 Conservation Across Human and Elephant Sequences Independent of Mutation Hotspots

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ABSTRACT

Large-bodied, long-lived species such as elephants exhibit lower-than-expected cancer incidence, a phenomenon described as Peto's paradox. One proposed explanation is the expansion of TP53 gene copies, which may enhance tumor suppressive capacity. Although mutation hotspots in human TP53 are well characterized, their evolutionary conservation across species remains insufficiently explored.

Here, residue-level conservation of six TP53 hotspot positions (R175, G245, R248, R249, R273, and R282) was evaluated using comparative sequence analysis of human and elephant TP53 proteins. Conservation scores were calculated and compared with residues from the DNA-binding domain using non-parametric statistical methods. Additionally, conservation patterns across the entire protein were examined.

Hotspot residues displayed high conservation (mean = 0.823), but no statistically significant difference was observed compared to non-hotspot residues ($p = 0.734$). Broader analysis revealed greater variability across the full protein sequence than within the DNA-binding domain. Structural mapping confirmed that hotspot residues are localized within this domain.

These findings suggest that conservation in TP53 is governed by domain-level constraints rather than hotspot-specific evolutionary pressure.

Keywords

TP53; mutation hotspots; comparative genomics; evolutionary conservation; DNA-binding domain

1. INTRODUCTION

The TP53 gene encodes a transcription factor that plays a central role in maintaining genomic integrity by regulating pathways such as DNA repair, apoptosis, and cell cycle control [1,2]. Alterations in TP53 are widely observed across diverse human cancers, with mutations frequently occurring at specific residues within the DNA-binding domain [3–5].

Across species, cancer incidence does not correlate directly with body size or lifespan. This observation, referred to as Peto's paradox [6], is particularly evident in large mammals such as elephants. Previous studies have reported an increased number of TP53 gene copies in elephants, including retrogenes, which may contribute to enhanced tumor suppression mechanisms [7–9].

While mutation hotspots in TP53 have been extensively studied in human cancers, their evolutionary conservation across species remains less clearly defined. In particular, it is unclear whether these hotspot residues are uniquely conserved compared to other residues within functionally important regions of the protein.

The present study addresses this question by performing a comparative sequence and structural analysis of TP53 across human and elephant datasets, with a focus on residue-level conservation within the DNA-binding domain.

2. MATERIALS AND METHODS

2.1 Study Design and Workflow

This study was designed to evaluate whether mutation hotspot residues in TP53 exhibit distinct evolutionary conservation compared to other residues within the protein. A computational workflow was developed integrating sequence retrieval, multiple sequence alignment, residue-level conservation analysis, statistical comparison, and structural visualisation.

2.2 Sequence Retrieval and Dataset Construction

The canonical human TP53 protein sequence was obtained from UniProt and NCBI databases [14]. Elephant TP53 sequences, including canonical and retrogene variants, were retrieved from Ensembl and NCBI [14].

2.3 Multiple Sequence Alignment

Protein sequences were aligned using a global alignment strategy to ensure consistent positional comparison across homologous residues. Standard alignment tools such as Clustal Omega [11] or MAFFT [12] were used for sequence alignment.

2.4. Identification and Mapping of Mutation Hotspots

Six TP53 mutation hotspots (R175, G245, R248, R249, R273, and R282) were selected based on established cancer mutation databases. These residues were mapped onto the aligned sequences using human TP53 numbering as a reference.

2.5. Residue-level Conservation Analysis

To quantify evolutionary conservation, a position-specific conservation score was calculated as the proportion of identical residues across aligned sequences. This allowed direct comparison of conservation levels across different positions in the protein.

Selection of Comparative Residue Groups

To evaluate whether hotspot residues exhibit unique conservation patterns, a comparative framework was established:

- a. **Hotspot residues** ($n = 6$)
- b. **Non-hotspot residues within the DNA-binding domain** ($n = 20$)
- c. **All residues across the TP53 sequence**

Non-hotspot residues were sampled from the DNA-binding domain (positions 100–300) to ensure fair comparison within a functionally constrained region.

2.6. Statistical Analysis

Differences in conservation between residue groups were evaluated using non-parametric statistical tests, including the Mann–Whitney U test and Kruskal–Wallis test. These non-parametric methods were chosen due to small sample sizes and non-normal data distribution.

2.7. Structural Mapping of Hotspot Residues

Structural mapping of hotspot residues was performed using the TP53 DNA-binding domain structure (PDB ID: 1TUP), and visualisation was carried out using PyMOL [\[17\]](#).

3. RESULTS

3.1. Conservation of TP53 Hotspot Residues

Residue-level conservation analysis across aligned TP53 sequences demonstrated that all six canonical hotspot residues—R175, G245, R248, R249, R273, and R282—are highly conserved across species. Conservation scores ranged from 0.688 to 1.000, with a mean conservation score of 0.823. These findings indicate that hotspot residues are located within evolutionarily constrained regions of the TP53 protein.

3.2. Comparison Between Hotspot and Domain-specific Non-hotspot Residues

To determine whether hotspot residues exhibit distinct conservation patterns, their conservation scores were compared with those of non-hotspot residues sampled from the DNA-binding domain. Non-hotspot residues showed comparable conservation levels, with a mean conservation score of **0.865**. Statistical analysis using the Mann–Whitney U test revealed **no significant difference** between hotspot and non-hotspot residues ($p = 0.734$). This result

indicates that conservation is not uniquely associated with mutation hotspots but is characteristic of the broader domain.

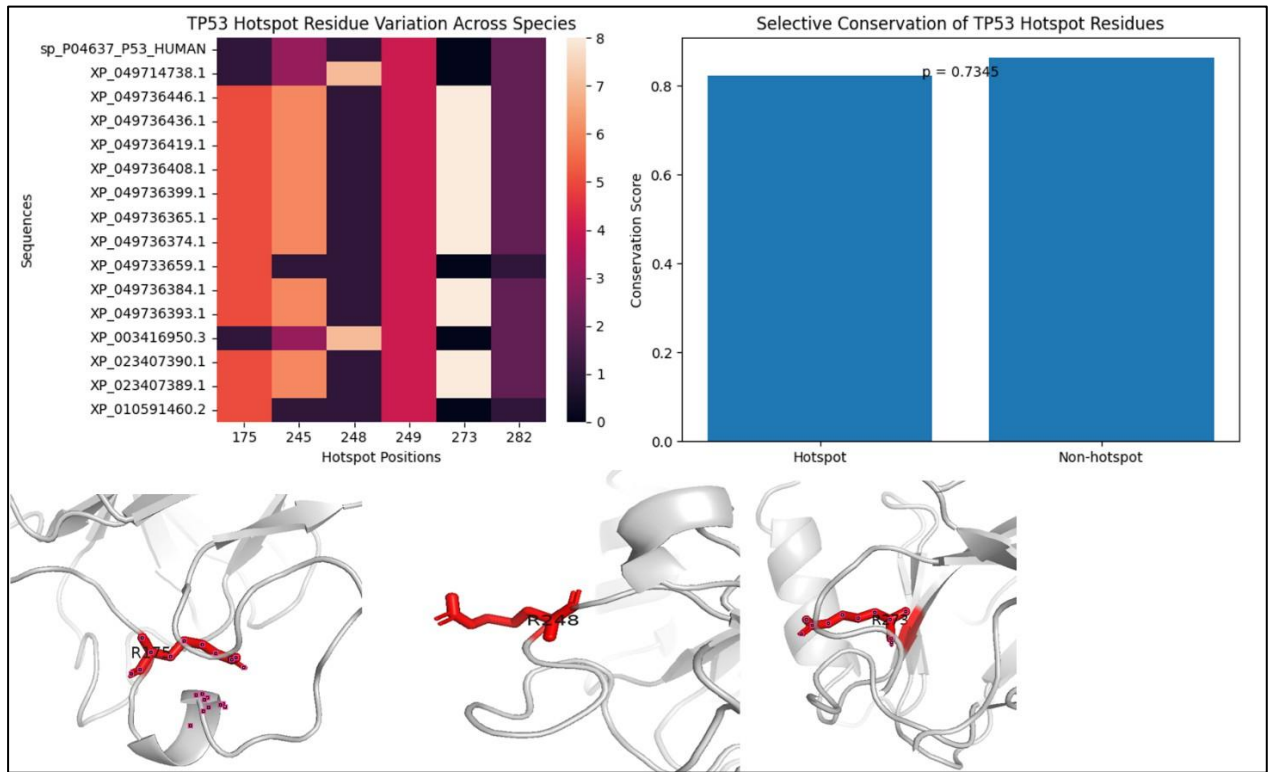


Figure 1. Comparative and structural analysis of TP53 hotspot conservation. (A) Heatmap showing residue variation across TP53 hotspot positions. (B) Comparison of conservation scores between hotspot and non-hotspot residues ($p = 0.734$). (C–E) Structural localization of representative hotspot residues R175, R248, and R273.

3.3. Distribution of Conservation Across TP53

To further evaluate conservation patterns, distribution-level analysis was carried out across three groups: hotspot residues, domain-specific non-hotspot residues, and the full TP53 sequence. Both hotspot and non-hotspot residues exhibited overlapping conservation distributions, indicating similar levels of evolutionary constraint within the DNA-binding domain. In contrast, conservation scores across the entire TP53 protein showed greater variability, including lower conservation outliers. These findings demonstrate that evolutionary constraint is primarily **domain-specific rather than hotspot-specific**.

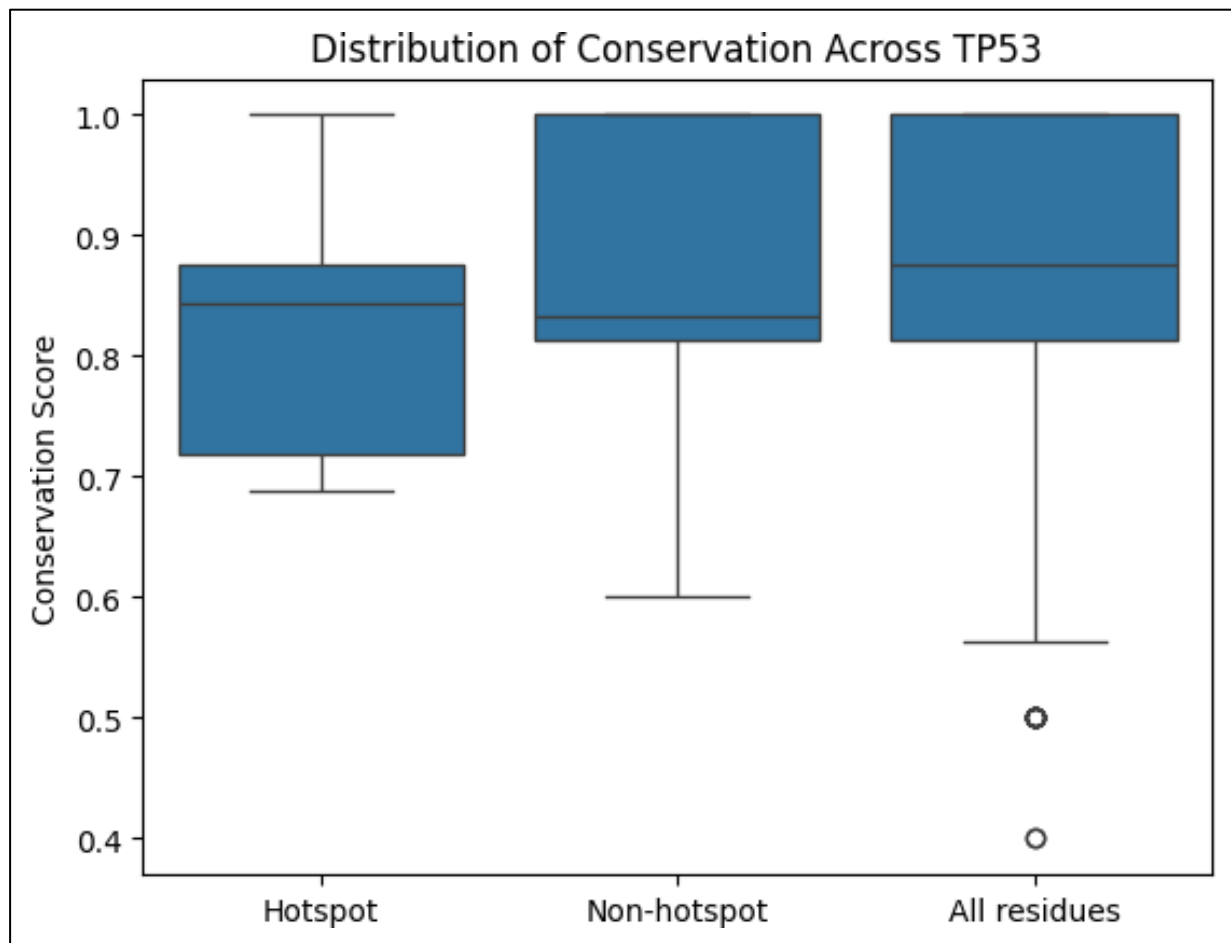


Figure 2. Distribution of conservation scores across TP53 residues. Boxplot showing conservation score distributions for hotspot residues, non-hotspot residues within the DNA-binding domain, and the full TP53 sequence. Overlapping distributions between hotspot and non-hotspot residues indicate comparable conservation levels, while the full protein exhibits greater variability.

3.4. Structural Localisation of Hotspot Residues

Structural mapping of hotspot residues onto the TP53 DNA-binding domain revealed that all hotspot residues are localised within this functional region. Representative residues (R175, R248, and R273) were visualised individually to illustrate their spatial positioning. This structural localisation supports the observation that hotspot residues are embedded within a highly conserved functional domain.

3.5. Phylogenetic Analysis of TP53 Sequences

Phylogenetic analysis revealed clear clustering of TP53 sequences according to species. The human TP53 sequence formed a distinct branch, while elephant TP53 sequences grouped separately. Elephant TP53 retrogene variants clustered together, indicating evolutionary divergence following gene duplication events.

This clustering provides an evolutionary framework for interpreting conservation patterns observed in the DNA-binding domain.

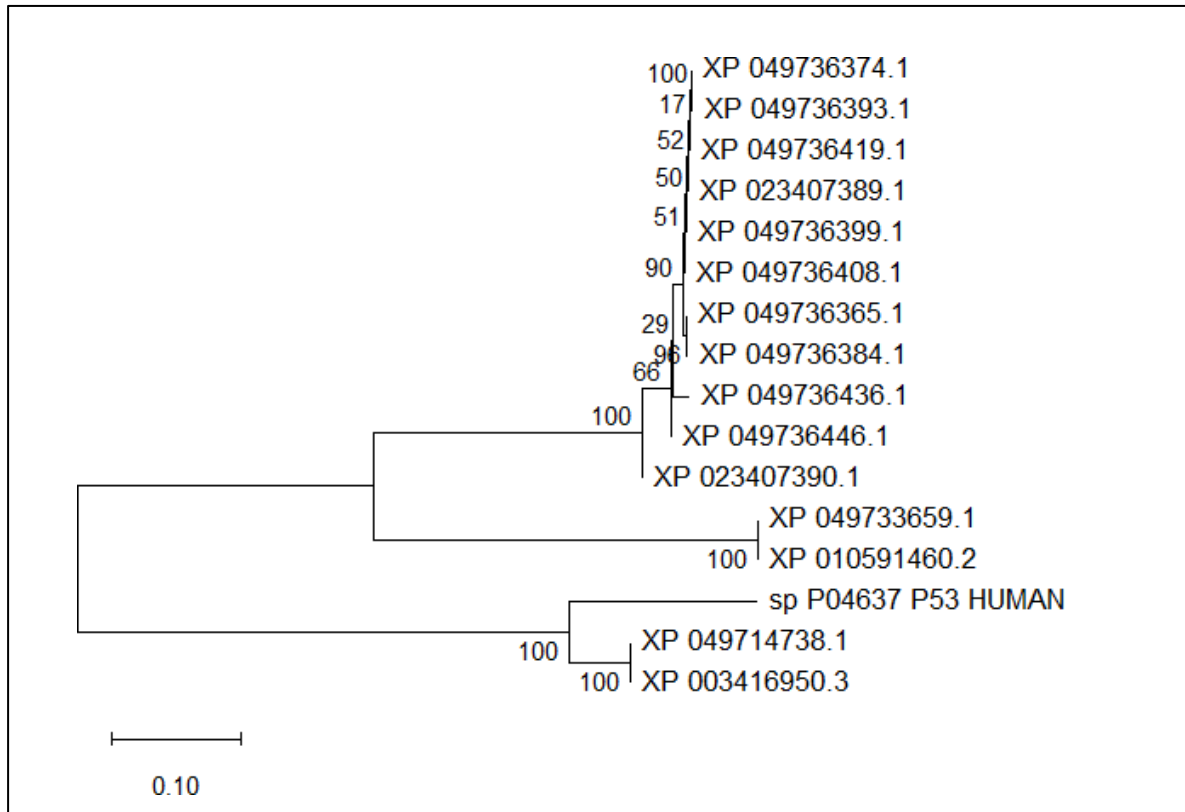


Figure 3. Phylogenetic relationships of TP53 sequences.

The phylogenetic tree shows clustering of human TP53 and elephant TP53 sequences, with retrogene variants forming distinct groups.

4. DISCUSSION

The present study aimed to evaluate whether TP53 mutation hotspot residues exhibit distinct evolutionary conservation compared to other residues within the protein. TP53 has been extensively studied as a critical tumour suppressor gene, with mutation hotspots well documented in cancer biology [3,4]. However, the evolutionary significance of these hotspot residues has remained less clear.

Although hotspot residues showed high conservation across species, similar conservation levels were observed for non-hotspot residues within the same domain. Statistical analysis confirmed that the difference between these groups was not significant. This indicates that evolutionary constraint in TP53 is not restricted to mutation-prone residues but extends across the entire DNA-binding domain.

These findings align with previous studies suggesting that functional domains within TP53 are subject to strong evolutionary constraints due to their role in DNA binding and transcriptional regulation [2].

The structural mapping of hotspot residues reinforces this interpretation. All analysed hotspot residues were localised within the DNA-binding domain, indicating that their functional importance is tied to domain integrity rather than isolated residue-specific conservation. These residues occupy structurally relevant positions, which may contribute to their sensitivity to mutation.

The lack of differential conservation between hotspot and non-hotspot residues suggests that mutation hotspots in TP53 are not defined by evolutionary conservation alone. Instead, these residues may represent positions that are structurally or functionally vulnerable. In this context, mutation frequency observed in cancer may reflect biochemical sensitivity or structural constraints rather than selective evolutionary pressure.

Phylogenetic analysis provides additional context by demonstrating clear separation between human TP53 and elephant TP53 sequences, with elephant retrogene variants forming distinct clusters. This supports the presence of gene duplication events and evolutionary divergence in elephant TP53. However, despite these differences, conservation patterns within the DNA-binding domain remain consistent, further emphasising the importance of domain-level constraints.

These findings emphasise that mutation susceptibility in TP53 is likely driven by structural and functional sensitivity rather than differential evolutionary conservation.

5. LIMITATIONS

This study is based on sequence-level conservation analysis and does not incorporate structural stability calculations, energetic modelling, or experimental validation. The conservation scoring approach relies on multiple sequence alignment and may be influenced by alignment quality and sequence selection. Additionally, the relatively small number of hotspot residues limits statistical power. Future studies integrating structural bioinformatics, larger datasets, and experimental assays are required to further elucidate the mechanistic basis of TP53 mutation susceptibility.

6. CONCLUSION

Here, comparative analysis of TP53 sequences across human and elephant revealed that mutation hotspot residues are not uniquely conserved relative to other residues within the DNA-binding domain. Instead, conservation is a domain-level property, reflecting the functional importance of this region.

These findings suggest that mutation hotspots in TP53 arise from structural and functional sensitivity rather than differential evolutionary conservation. This work provides a computational framework for integrating sequence, statistical, and structural analysis to better understand mutation patterns in cancer-related genes.

7. AUTHOR CONTRIBUTIONS

Ms Ritika Rajendra. Rawat conceptualised the study, performed sequence analysis, conducted statistical analysis, generated figures, and wrote the manuscript. Mrs Sermarani Nadar contributed to data interpretation and methodology refinement. Dr Gursimran Kaur Uppal contributed to the manuscript review and supervision.

8. FUNDING

No external funding was received.

9. CONFLICT OF INTEREST

The authors declare no conflict of interest.

10. DATA AVAILABILITY

All sequence data used here were obtained from publicly available databases, including NCBI, UniProt, and Ensembl. The multiple sequence alignment file and analysis scripts are available upon reasonable request.

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